MSG No. 163.84 $P_c\bar{3}1c$ [Type IV, trigonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3.m'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,7,8,9,16,17,18,22,23,24]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,5,6,10,11,12,13,14,15,19,20,21]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3'.m'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 3: Wyckoff site: `4c`, site symmetry: `3.2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,2,3,16,17,18]`
2	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[4,5,6,13,14,15]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[7,8,9,22,23,24]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[10,11,12,19,20,21]`

Table 4: Wyckoff site: `4d`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[7,8,9,10,11,12]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[13,14,15,16,17,18]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[19,20,21,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `3.m'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,22,23,24]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4,5,6,19,20,21]`
3	$[0,\ 0,\ -z]$	`[7,8,9,16,17,18]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[10,11,12,13,14,15]`

Table 6: Wyckoff site: `6f`, site symmetry: `..2'/m'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,7,17,23]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,8,18,24]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,9,16,22]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[4,10,15,21]
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,11,13,19]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,12,14,20]

Table 7: Wyckoff site: 6g, site symmetry: ..2/m'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[1,5,19,23]
2	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[2,6,20,24]
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[3,4,21,22]
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[7,11,13,17]
5	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[8,12,14,18]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[9,10,15,16]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,2,3]
2	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	[4,5,6]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,8,9]
4	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	[10,11,12]
5	$\left[\frac{1}{3}, \frac{2}{3}, z + \frac{1}{2}\right]$	[13,14,15]
6	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[16,17,18]
7	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2} - z\right]$	[19,20,21]
8	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[22,23,24]

Table 9: Wyckoff site: 12i, site symmetry: ..2'

No.	position	mapping
1	$[x, -x, 0]$	[1,18]
2	$[x, 2x, 0]$	[2,16]
3	$[-2x, -x, 0]$	[3,17]
4	$\left[x, 2x, \frac{1}{2}\right]$	[4,14]
5	$\left[-2x, -x, \frac{1}{2}\right]$	[5,15]
6	$\left[x, -x, \frac{1}{2}\right]$	[6,13]
7	$[-x, x, 0]$	[7,24]
8	$[-x, -2x, 0]$	[8,22]
9	$[2x, x, 0]$	[9,23]
10	$\left[-x, -2x, \frac{1}{2}\right]$	[10,20]
11	$\left[2x, x, \frac{1}{2}\right]$	[11,21]
12	$\left[-x, x, \frac{1}{2}\right]$	[12,19]

Table 10: Wyckoff site: `12j`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\, -x,\, \frac{1}{4}\right]$	`[1,6]`
2	$\left[x,\, 2x,\, \frac{1}{4}\right]$	`[2,4]`
3	$\left[-2x,\, -x,\, \frac{1}{4}\right]$	`[3,5]`
4	$\left[-x,\, x,\, \frac{3}{4}\right]$	`[7,12]`
5	$\left[-x,\, -2x,\, \frac{3}{4}\right]$	`[8,10]`
6	$\left[2x,\, x,\, \frac{3}{4}\right]$	`[9,11]`
7	$\left[x,\, -x,\, \frac{3}{4}\right]$	`[13,18]`
8	$\left[x,\, 2x,\, \frac{3}{4}\right]$	`[14,16]`
9	$\left[-2x,\, -x,\, \frac{3}{4}\right]$	`[15,17]`
10	$\left[-x,\, x,\, \frac{1}{4}\right]$	`[19,24]`
11	$\left[-x,\, -2x,\, \frac{1}{4}\right]$	`[20,22]`
12	$\left[2x,\, x,\, \frac{1}{4}\right]$	`[21,23]`

Table 11: Wyckoff site: `12k`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\, 0,\, z]$	`[1,23]`
2	$[0,\, x,\, z]$	`[2,24]`
3	$[-x,\, -x,\, z]$	`[3,22]`
4	$\left[x,\, x,\, \frac{1}{2}-z\right]$	`[4,21]`
5	$\left[-x,\, 0,\, \frac{1}{2}-z\right]$	`[5,19]`
6	$\left[0,\, -x,\, \frac{1}{2}-z\right]$	`[6,20]`
7	$[-x,\, 0,\, -z]$	`[7,17]`
8	$[0,\, -x,\, -z]$	`[8,18]`
9	$[x,\, x,\, -z]$	`[9,16]`
10	$\left[-x,\, -x,\, z+\frac{1}{2}\right]$	`[10,15]`
11	$\left[x,\, 0,\, z+\frac{1}{2}\right]$	`[11,13]`
12	$\left[0,\, x,\, z+\frac{1}{2}\right]$	`[12,14]`

Table 12: Wyckoff site: `24l`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x-y,\, z]$	`[2]`
3	$[-x+y,\, -x,\, z]$	`[3]`
4	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	`[4]`
5	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	`[5]`
6	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	`[6]`
7	$[-x,\, -y,\, -z]$	`[7]`
8	$[y,\, -x+y,\, -z]$	`[8]`
9	$[x-y,\, x,\, -z]$	`[9]`
10	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	`[10]`
11	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	`[11]`
12	$\left[y,\, x,\, z+\frac{1}{2}\right]$	`[12]`

continued ...

Table 12

No.	position	mapping
13	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[13]
14	$\left[-y,\, x-y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[-x+y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$[x,\, x-y,\, -z]$	[16]
17	$[-x+y,\, y,\, -z]$	[17]
18	$[-y,\, -x,\, -z]$	[18]
19	$\left[-x,\, -y,\, \frac{1}{2}-z\right]$	[19]
20	$\left[y,\, -x+y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[x-y,\, x,\, \frac{1}{2}-z\right]$	[21]
22	$[-x,\, -x+y,\, z]$	[22]
23	$[x-y,\, -y,\, z]$	[23]
24	$[y,\, x,\, z]$	[24]